

Started on 11/05/2026

Built on wokwi.com

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Smart Light

This project is a simulation of an energy-conscious SmartLight that changes light output depending on the amount of sunlight, motion detection, and battery level. This project is built on wokwi.com. [Wokwi.com](https://wokwi.com) is a browser-based electronics simulator where you can build and test digital circuits virtually. This project is done digitally to reduce costs and enhance exposition.

When there is no sunlight and no motion detected, the SmartLight LED will remain on but at a lower brightness. When there is no sunlight detected and motion is detected, the SmartLight LED will be on at full brightness. When there is low battery, the “low battery” LED will be flashing. If the SmartLight is on while there is low battery, the SmartLight LED will have a lower brightness. There will also be a third LED simulating the sun.

Table 1. The table below shows the logic behind the SmartLight and the Low Battery LED.

Condition	SmartLight	Low Battery LED	Sunlight LED
Sunlight Detected	Off	Off	On
No Sunlight Detected	On	Off	Off
No Sunlight + Low Battery	Low brightness	Flashing	Off
No sunlight + Low battery + Motion detected	Medium brightness	Flashing	Off

Notes:

Originally, the project would have had three switches. One would toggle the SmartLight day/night modes, the second one would toggle the motion-detected switch, and the third one would toggle the low battery mode. This was changed for a more realistic approach using a photoresistor, a motion sensor, and a potentiometer to simulate the different scenarios. All progress from Figure 2 was lost due to a website malfunction. Redone as shown in Figure 3. Since the potentiometer is simulating the battery, and it ranges from 0 to 1023, “low battery” will be 15% of this, which is around 153.

For the first test, as seen in Figure 5, there were some issues with the low battery mode. As the low battery LED did not flash during nighttime. The hardest part to code was probably the timing. When 10 seconds had passed since the last motion detected, the light dimmed back down. So a timer was needed to count those 10 seconds. Resistors were added as shown in Figure 7 to make it more realistic.

Connections Organization

Power & Ground Standards

- Ground (GND): Brown wire (Used universally for all ground connections)
- 5V Power Rail: Light Blue wire

Input Sensors

- Digital Input 2 (D2): Pink wire - Motion Sensor (PIR)
- Digital Input 3 (D3): Purple wire - Photoresistor (LDR Module)
- Analog Input 0 (A0): Blue wire - Potentiometer ("Low Battery" simulator)

Output Components (LEDs)

Note: All LED cathode/ground returns use the global standard green or brown wire configuration, while their individual anode/signal lines are color-coded as follows:

- Digital PWM Pin 9 (D9): White wire - SmartLight LED
- Digital PWM Pin 10 (D10): Yellow wire - Sun Status LED
- Digital PWM Pin 11 (D11): Red wire - Low Battery LED

Pictures documenting progress:

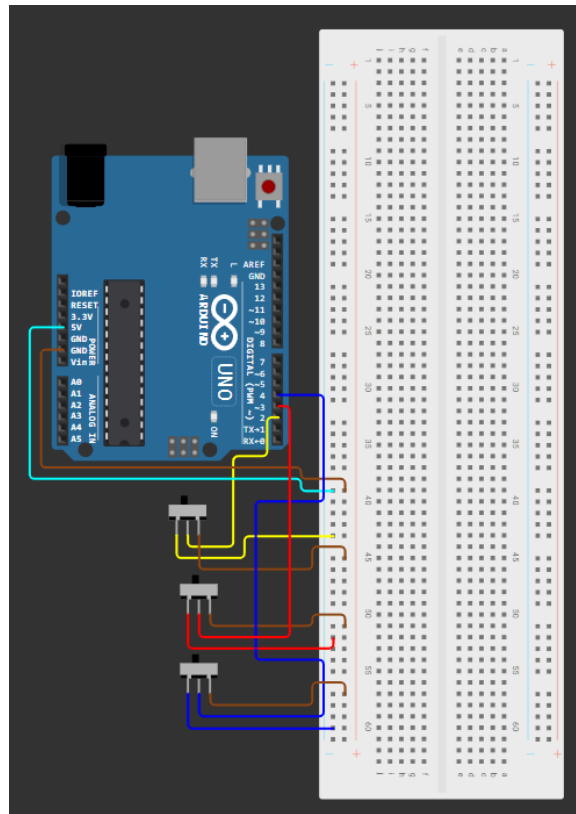


Figure 1. Original Switch-based Project

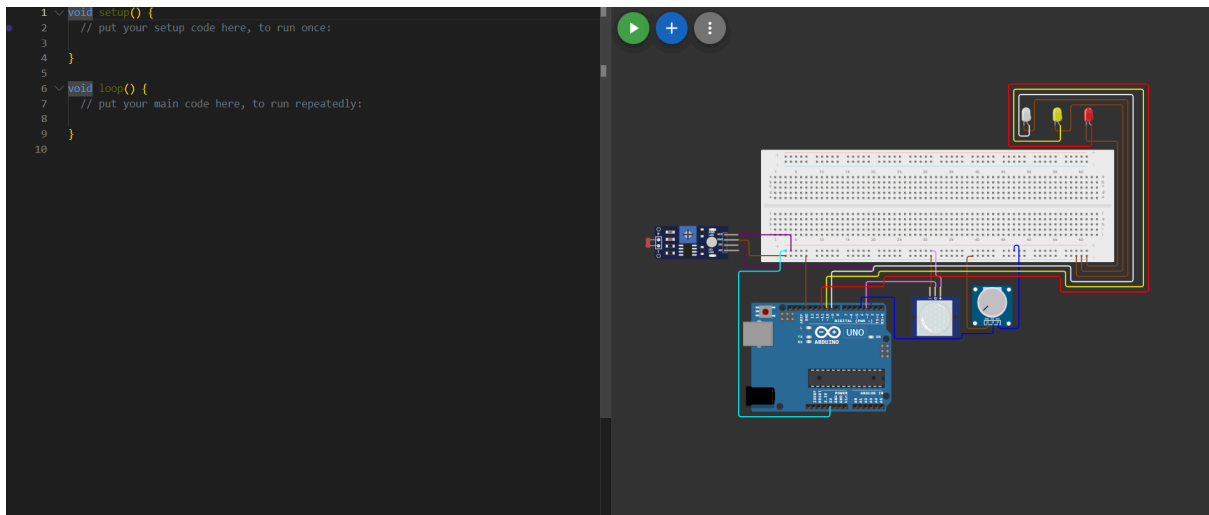


Figure 2. Lost diagram for the project.

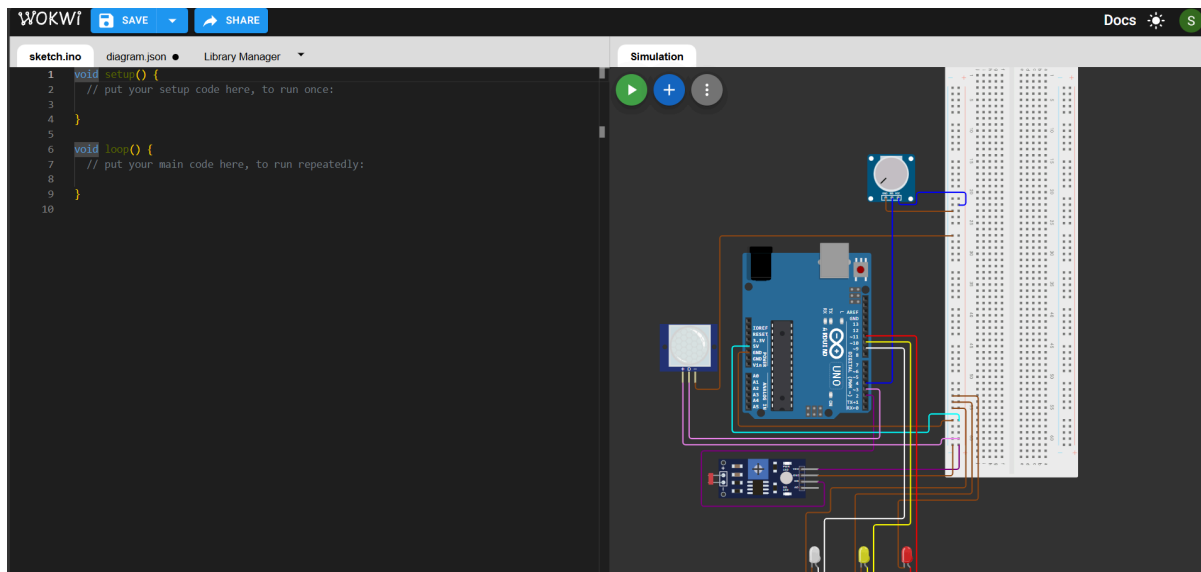


Figure 3. Redone circuit for Figure 2.

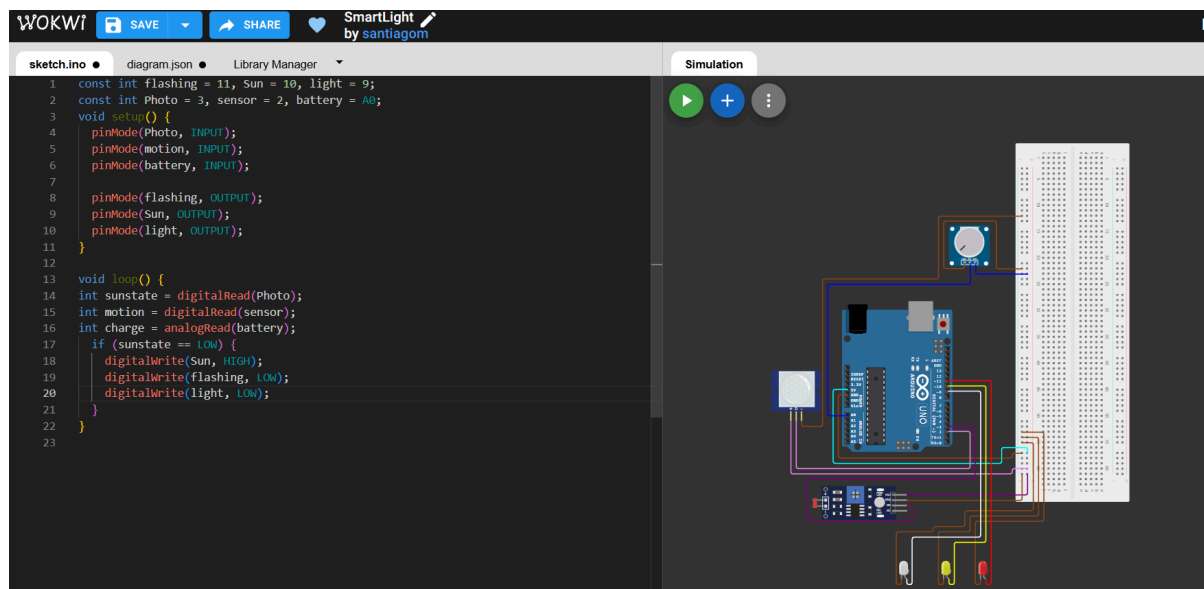


Figure 4. Progress in coding.

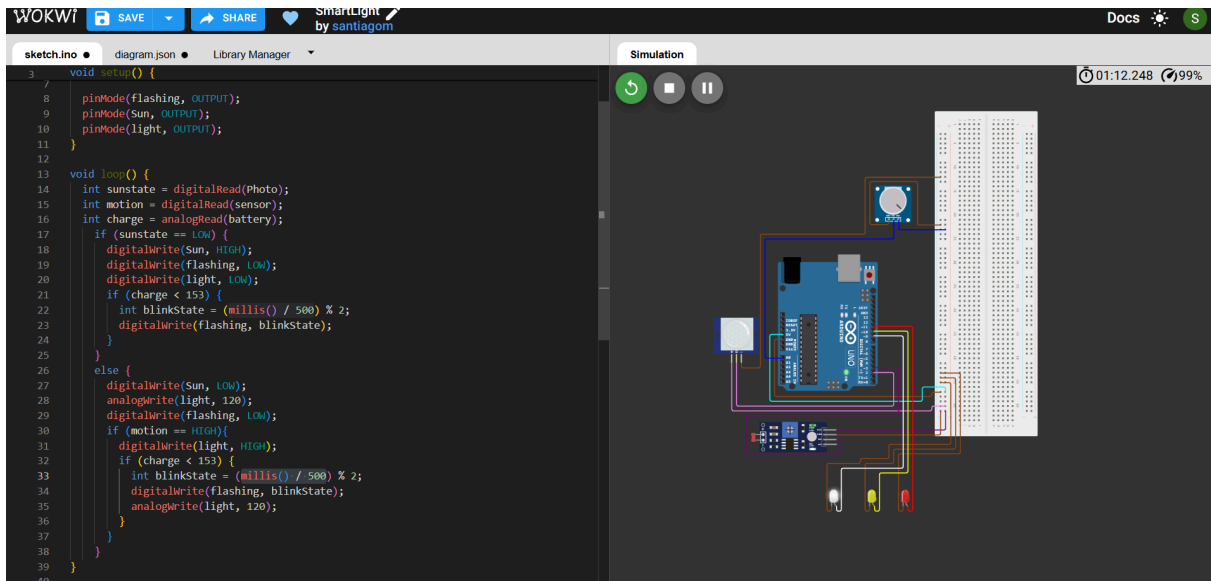


Figure 5. First Test of Project.

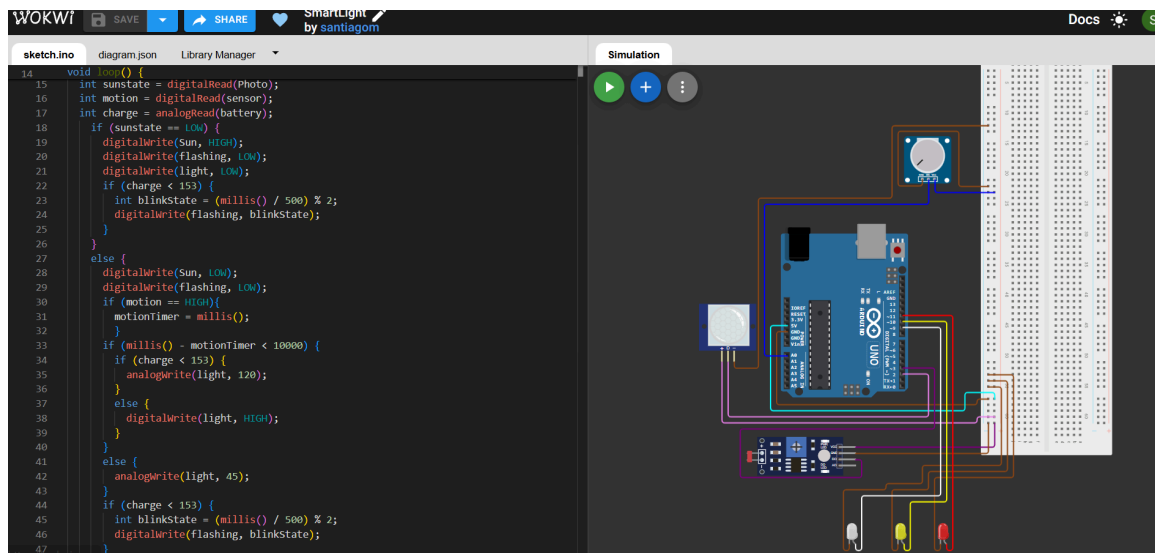


Figure 6. Finished Project

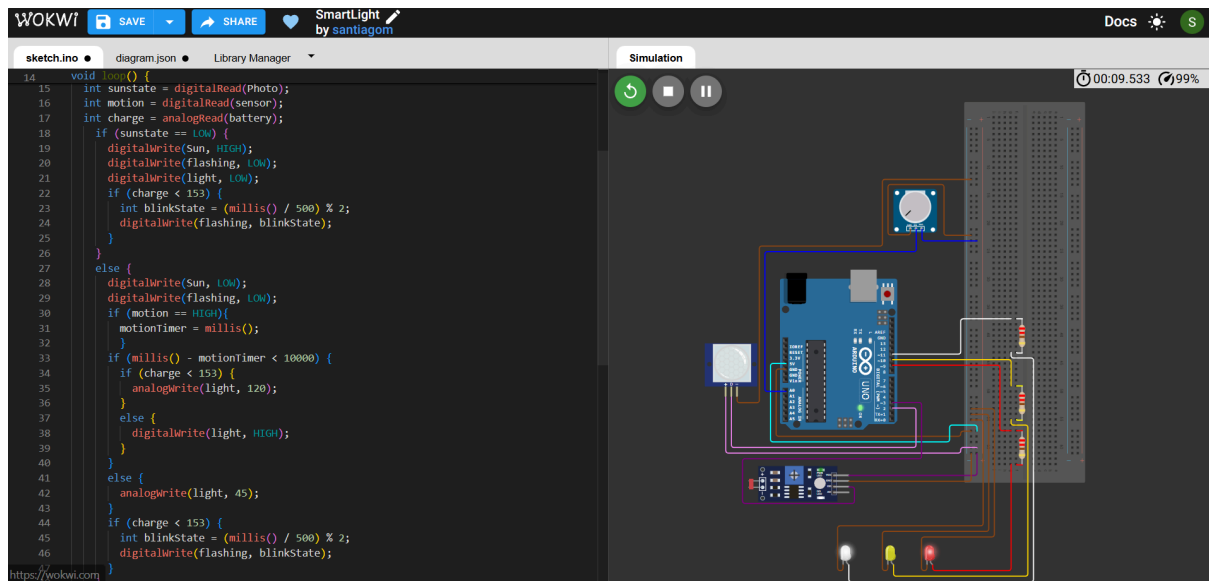


Figure 7. Redone wiring with resistors to make it more realistic.

The code used in the SmartLight is shown below:

```
const int flashing = 11, Sun = 10, light = 9;
const int Photo = 3, sensor = 2, battery = A0;
unsigned long motionTimer = 0;
void setup() {
    pinMode(Photo, INPUT);
    pinMode(sensor, INPUT);
    pinMode(battery, INPUT);

    pinMode(flashing, OUTPUT);
    pinMode(Sun, OUTPUT);
    pinMode(light, OUTPUT);
}

void loop() {
    int sunstate = digitalRead(Photo);
    int motion = digitalRead(sensor);
    int charge = analogRead(battery);
    if (sunstate == LOW) {
        digitalWrite(Sun, HIGH);
        digitalWrite(flashing, LOW);
        digitalWrite(light, LOW);
        if (charge < 153) {
            int blinkState = (millis() / 500) % 2;
            digitalWrite(flashing, blinkState);
        }
    }
    else {
        digitalWrite(Sun, LOW);
        digitalWrite(flashing, LOW);
        if (motion == HIGH) {
            motionTimer = millis();
        }
        if (millis() - motionTimer < 10000) {
            if (charge < 153) {
                analogWrite(light, 120);
            }
            else {
                digitalWrite(light, HIGH);
            }
        }
        else {
            analogWrite(light, 45);
        }
        if (charge < 153) {
            int blinkState = (millis() / 500) % 2;
            digitalWrite(flashing, blinkState);
        }
    }
}
```


This project was completed on 05/20/2026. To see the full project, go to

<https://wokwi.com/projects/464054451096978433>

This project has reinforced my knowledge of Arduino basics, computer programming, and electrical components. It has also taught me about new electrical components I did not previously know about, such as a photoresistor, a PIR, and a potentiometer. This was a very rewarding project, and I look forward to doing similar things in the future, building on my current knowledge.

“For we are His workmanship, created in Christ Jesus for good works, which God prepared beforehand that we should walk in them”

(Ephesians 2:10)